

Mendelian inheritance has its physical basis in the behavior of chromosomes

When Gregor Mendel proposed the existence of “hereditary factors” in 1860, no cellular structures had been identified that could house these imaginary units, and most biologists were skeptical. When the processes of mitosis and meiosis were worked out later that century, biologists saw parallels between the behavior of Mendel’s proposed factors (genes) during sexual life cycles and the behavior of chromosomes: As shown in **Figure 15.1**, chromosomes and genes are both present in pairs in diploid cells, and homologous chromosomes separate and alleles segregate during the process of meiosis. After meiosis, fertilization restores the paired condition for both chromosomes and genes. Around 1902, Walter S. Sutton, Theodor Boveri, and others independently noted these parallels and began to develop the **chromosome theory of inheritance**. According to this theory, Mendelian genes have specific loci (sites) along chromosomes, and it is the chromosomes that undergo segregation and independent assortment.

The first solid evidence associating a specific gene with a specific chromosome came early in the 1900s from the work of Thomas Hunt Morgan, an experimental embryologist at Columbia University. Although Morgan was initially doubtful about both Mendelian genetics and the chromosome theory, his early experiments provided convincing evidence that chromosomes are indeed the location of Mendel’s heritable factors.

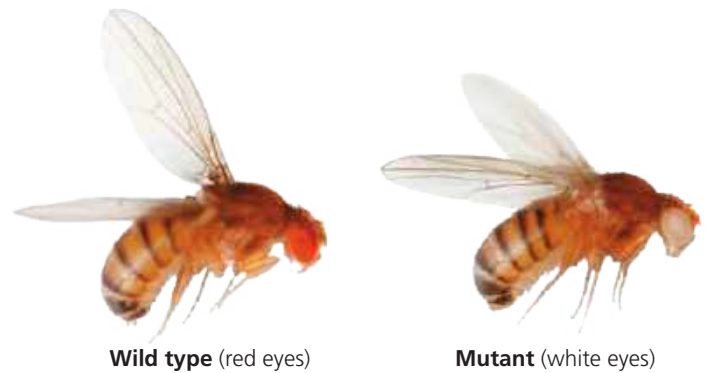
Morgan’s Choice of Experimental Organism

Many times in the history of biology, important discoveries have come to those insightful or lucky enough to choose an experimental organism suitable for the research problem being tackled. Mendel chose the garden pea because a number of distinct varieties were available. For his work, Morgan selected a species of fruit fly, *Drosophila melanogaster*, a common insect that feeds on the fungi growing on fruit. Fruit flies are prolific breeders; a single mating will produce hundreds of offspring, and a new generation can be bred every two weeks. Morgan’s laboratory began using this convenient organism for genetic studies in 1907 and soon became known as “the fly room.”

Another advantage of the fruit fly is that it has only four pairs of chromosomes, which are easily distinguishable with a light microscope. There are three pairs of autosomes and one pair of sex chromosomes. Female fruit flies have a pair of homologous X chromosomes, and males have one X chromosome and one Y chromosome.

While Mendel could readily obtain different pea varieties from seed suppliers, Morgan was probably the first person to want different varieties of the fruit fly. He faced the tedious task of carrying out many matings and then microscopically

▼ **Figure 15.2 Morgan’s first mutant.** Wild-type *Drosophila* flies have red eyes (left). Among his flies, Morgan discovered a mutant male with white eyes (right). This variation made it possible for Morgan to trace a gene for eye color to a specific chromosome.



inspecting large numbers of offspring in search of naturally occurring variant individuals. After many months of this, he complained, “Two years’ work wasted. I have been breeding those flies for all that time and I’ve got nothing out of it.” Morgan persisted, however, and was finally rewarded with the discovery of a single male fly with white eyes instead of the usual red. The phenotype for a character most commonly observed in natural populations, such as red eyes in *Drosophila*, is called the **wild type** (**Figure 15.2**). Traits that are alternatives to the wild type, such as white eyes in *Drosophila*, are called *mutant phenotypes* because they are due to alleles assumed to have originated as changes, or mutations, in the wild-type allele.

Morgan and his students invented a notation for symbolizing alleles in *Drosophila* that is still widely used for fruit flies. For a given character in flies, the gene takes its symbol from the first mutant (non-wild type) discovered. Thus, the allele for white eyes in *Drosophila* is symbolized by *w*. A superscript + identifies the allele for the wild-type trait: *w*⁺ for the allele for red eyes, for example. Over the years, a variety of gene notation systems have been developed for different organisms. For example, human genes are usually written in all capitals, such as *HTT* for the gene involved in Huntington’s disease. (More than one letter may be used for an allele, as in *HTT*.)

Correlating Behavior of a Gene’s Alleles with Behavior of a Chromosome Pair: Scientific Inquiry

Morgan mated his white-eyed male fly with a red-eyed female. All the F₁ offspring had red eyes, suggesting that the wild-type allele is dominant. When Morgan bred the F₁ flies to each other, he observed the classical 3:1 phenotypic ratio among the F₂ offspring. However, there was a surprising additional result: The white-eye trait showed up only in males. All the F₂ females had red eyes, while half the males had red eyes and half had white eyes. Therefore, Morgan concluded